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EE558 HW 1

1. a. 0.4 μm → $\nu = \frac{3e8}{0.4e\text{-}6} = 7.5\text{THz}$

   0.7μm → $\nu = \frac{3e8}{0.7e\text{-}6} = 4.3\text{THz}$

   b. (i) max duration

   $1\text{Mbit/sec} = \frac{1000000 \text{bits}}{\text{sec}} \rightarrow$ 1e-6 seconds /bit

   1 Gbit/sec → 1e-9 seconds/bit

   (ii) power level = 1mW

   1Gbit/sec → energy = 1mW × 1e-9 = <u>1e-12 J</u>

   1Mbit/sec → E = 1mW × 1e-6 = <u>1e-9 J</u>

   c. 10^5dB/km → 0.1dB/mm, 5mm → 0.5dB attenuation

   $P = 10^{(0.5/10)} =$ <u>89%</u> of the original light remains.

   d. $v = \frac{c}{n} = \frac{3e8}{1.5} = \boxed{2e8 \text{ m/s}}$

2. total attenuation of fiber: 0.5dB/km × 50km = 25dB

   splices: 0.3dB/splice × 16 splices = 4.8dB

   2 connectors : 2dB

   a. total loss in dB: 31.8dB

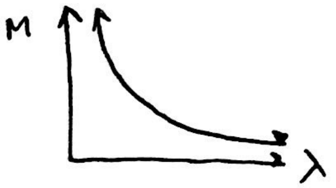
   b. power input to the receiver: $P = \frac{1\text{mW}}{10^{\left(\frac{-31.8\text{dB}}{10\text{dB}}\right)}} =$ <u>1.51 W</u>

3. # of modes in waveguide = M

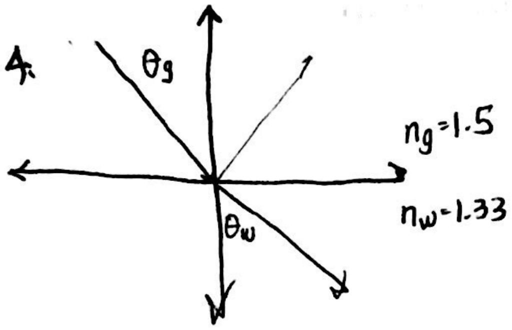
   a) $V = \frac{2\pi r}{\lambda}\sqrt{n_1^2 - n_2^2} = \frac{2\pi(100\mu m)}{1.3\mu m}\sqrt{1.49^2 - 1.48^2} = 83.29$

   $M = \frac{2V}{\pi} = 53$ modes

   b.



4.



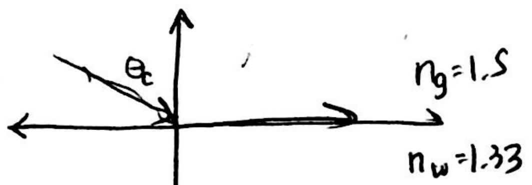
$\theta_g = 45^\circ$

$n_g \sin\theta_g = n_w \sin\theta_w$

$\sin\theta_w = \dfrac{1.5\sin(45^\circ)}{1.33}$

$\theta_w = \arcsin(0.797) = \underline{52.9^\circ}$

5.



$\theta_c = \arcsin\left(\dfrac{1.33}{1.5}\right) = 62.46^\circ$

The wave must come from the medium with the higher index of refraction.

6. a) $\lambda = 1.55\mu m$ signal $\rightarrow \nu = \dfrac{c}{\lambda} = \underline{193\,THz}$

b) 0.1nm bandwidth $\rightarrow \Delta\nu = \dfrac{c}{\Delta\lambda} = \underline{3\,EHz}$

7. a) energy of photon at $0.82\,\mu m$

$E = \dfrac{hc}{\lambda} = \dfrac{6.626e{-}34 \times 3e8}{0.82e{-}6} \times \dfrac{6.24e18\,eV}{1J} = \underline{1.513\,eV}$

$1.3um \rightarrow E = \underline{0.95eV}$

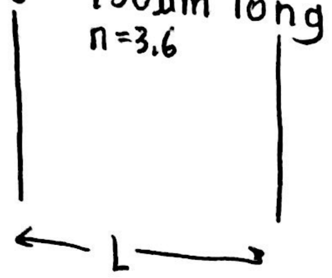
$1.55um \rightarrow E = \underline{0.8002eV}$

b. $1.24eV \rightarrow \lambda = \dfrac{hc}{E} = \dfrac{6.624e{-}34 \times 3e8}{1.24eV} \times \dfrac{6.24e18eV}{1J} = \underline{1\,\mu m}$

$f = \dfrac{c/n}{\lambda} = \dfrac{3e8/1.475}{1e{-}6} = \underline{203.4\,THz}$

8. cavity ⇒ 430 μm long

a)



$\Delta\lambda = 1.53\,\mu m - 1.57\,\mu m = 0.04\,\mu m$

$\rightarrow \Delta f = 36\,THz$

$2L = m\lambda = m\frac{c/n}{f}$

$m = \frac{2fLn}{c} \rightarrow \Delta m \approx \Delta f \cdot m'(f)$

$\Delta m \approx (36e{+}12)\left(\frac{2Ln}{c}\right)$

$\approx \underline{371\text{ modes}}$

b. $0.43\,\mu m \longrightarrow$ 0 modes

9. length transmitted for Ray B: L

length transmitted for Ray A: $\frac{L}{\sin\phi}$

$\sin\phi = \frac{n_2}{n_1}$ ⟶

$\Delta T = \dfrac{\frac{L}{\sin\phi} - L}{\frac{c}{n_1}}$ (time delay for tw rays)

$\Delta T = \cancel{n_1 L\left(\frac{n_1}{n_2}\right)}\ \frac{Ln_1\left(\frac{n_1}{n_2} - 1\right)}{c}$

$\Delta T/km = \boxed{\dfrac{n_1\left(\frac{n_1}{n_2} - 1\right)}{c}}$ (time delay per k

b. $\Delta T/km = 5.05e{-}11$ s delay/km

c. # of spatial modes = $\left(\frac{2(1.46)}{1.55e{-}6}\right)\sqrt{1.475^2 - 1.46^2}$

$= \underline{3.95e6\text{ spatial modes}}$